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24MA210

JSS MAHAVIDYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Mathematics

Second Semester B.E. Degree: Test-II

ENGINEERING MATHEMATICS-II

Duration: 1 Hour 15 Minutes | Date: 14/07/2025 | Max. Marks: 30

NOTE:
1. Answer all questions in **PART-A**.
2. Questions in **PART-B** have internal choice.

PART-A

Q.No.	CO	CD	PI	Question	Marks
1	CO-5	1	1.3.1	Evaluate: $\int_1^2 \int_1^3 y^2 \, dx \, dy$	05
2	CO-4	2	1.3.1	Using Taylor's series method find $y(0.1)$ considering upto third degree term if $y(x)$ satisfies the equation $\frac{dy}{dx} = x^2 + y^2$; $y(0) = 1$.	05
3	CO-4	2	1.3.1	Given that $\frac{dy}{dx} = x - y^2$ and the data $y(0) = 0,\ y(0.2) = 0.02,$ $y(0.4) = 0.0795,\ y(0.6) = 0.1762.$ Compute y at $x = 0.8$ by applying Milne's method.	05

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PART-B

Q.No.	CO	CD	PI	Question	Marks
4 a)	CO-4	2	1.1.1	Find a real root of the equation $x^3 + 5x - 11 = 0$ using Regula-Falsi method correct to four decimal places.	07
				OR	
4 b)	CO-4	2	1.1.1	Using Newton-Raphson method, find a real root of the equation $x^3 - 2x - 5 = 0$ correct to four decimal places.	07

Q.No.	CO	CD	PI	Question	Marks
5 a)	CO-5	1	1.1.1	Evaluate: $\int_0^1 \int_x^{\sqrt{x}} (xy)\ dy\ dx$	08
				OR	
5 b)	CO-5	1	1.1.1	Evaluate: $\int_1^2 \int_2^3 \int_3^4 (xyz)\ dz\ dy\ dx$	08

Course Outcome: At the end of the course the students will have the ability to	
CO-4:	Comprehend and apply numerical techniques to solve engineering problems.
CO-5:	Comprehend and apply the concepts of integral calculus.

Performance Indicators:	
1.1.1	Apply mathematical techniques such as calculus, linear algebra, and statistics to solve problems
1.3.1	Apply fundamental engineering concepts to solve engineering problems

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